Algorithm Sample 8

Dataset-expansion sample

1 Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

Algorithm 1 Delta Particle Filter

1. Input: data $y_{1:T}$, level $l \in \mathbb{N}$, particle number $N \in \mathbb{N}$ and parameter $\theta \in \Theta$.
2. Initialize: For $i \in \{1, \dots, N\}$, independently generate $\overline{W}^{i,l}_{\Delta_l:1}$ from $\mathcal{N}(0, \Delta_l)$. For $(i, k) \in \{1, \dots, N\} \times \{1, \dots, \Delta^{-1}_{l-1}\}$ set $\overline{W}^{i,l-1}_{k\Delta_{l-1}} = \overline{W}^{i,l}_{2k\Delta_l} + \overline{W}^{i,l}_{(2k-1)\Delta_l}$. Set $t = 1$, $\hat{\hat{p}}^N(y_{1:0}) = 1$ for convention and go to step 3.
3. Iterate: For $i \in \{1, \dots, N\}$ compute
$$\tilde{u}^i_t = \frac{\max\{\kappa_{t,l}(\overline{w}^{i,l}_{\Delta_l:t}), \kappa_{t,l-1}(\overline{w}^{i,l-1}_{\Delta_{l-1}:t})\}}{\sum_{j=1}^N \max\{\kappa_{t,l}(\overline{w}^{j,l}_{\Delta_l:t}), \kappa_{t,l-1}(\overline{w}^{j,l-1}_{\Delta_{l-1}:t})\}}.$$
Set $\hat{p}^N(y_{1:t}) = \hat{p}^N(y_{1:t-1})\frac{1}{N}\sum_{i=1}^N \max\{\kappa_{t,l}(\overline{w}^{i,l}_{\Delta_l:t}), \kappa_{t,l-1}(\overline{w}^{i,l-1}_{\Delta_{l-1}:t})\}$. Then sample $(\overline{w}^{1:N,l}_{\Delta_l:t}, \overline{w}^{1:N,l-1}_{\Delta_{l-1}:t})$ with replacement from $(\overline{w}^{1:N,l}_{\Delta_l:t}, \overline{w}^{1:N,l-1}_{\Delta_{l-1}:t})$ using probabilities $\tilde{u}^{1:N}_t$. For $i \in \{1, \dots, N\}$, independently generate $\overline{W}^i_{t+\Delta_l:t+1}$ from $\mathcal{N}(0, \Delta_l)$. For $(i, k) \in \{1, \dots, N\} \times \{1, \dots, \Delta^{-1}_{l-1}\}$ set $\overline{W}^{i,l-1}_{t+k\Delta_{l-1}} = \overline{W}^{i,l}_{t+2k\Delta_l} + \overline{W}^{i,l}_{t+(2k-1)\Delta_l}$. Set $t = t + 1$ and if $t = T$ go to step 4, otherwise restart step 3.
4. Grand Selection: For $i \in \{1, \dots, N\}$ compute $\tilde{u}^i_T = \frac{\max\{\kappa_{T,l}(\overline{w}^{i,l}_{\Delta_l:T}), \kappa_{T,l-1}(\overline{w}^{i,l-1}_{\Delta_{l-1}:T})\}}{\sum_{j=1}^N \max\{\kappa_{T,l}(\overline{w}^{j,l}_{\Delta_l:T}), \kappa_{T,l-1}(\overline{w}^{j,l-1}_{\Delta_{l-1}:T})\}}$. Set $\hat{p}^N(y_{1:T}) = \hat{p}^N(y_{1:T-1})\frac{1}{N}\sum_{i=1}^N \max\{\kappa_{T,l}(\overline{w}^{i,l}_{\Delta_l:T}), \kappa_{T,l-1}(\overline{w}^{i,l-1}_{\Delta_{l-1}:T})\}$. Sample one $(\overline{w}^l_{\Delta_l:T}, \overline{w}^{l-1}_{\Delta_{l-1}:T})$ from $(\overline{w}^{1:N,l}_{\Delta_l:T}, \overline{w}^{1:N,l-1}_{\Delta_{l-1}:T})$ using $\tilde{u}^{1:N}_T$ and go to step 5.
5. Output: trajectories $(\overline{w}^l_{\Delta_l:T}, \overline{w}^{l-1}_{\Delta_{l-1}:T})$ and normalizing constant estimate $\hat{\hat{p}}^N(y_{1:T})$.